

Edge Score for Fixing1 vs Fixing2 Arbitrage - Specification

Objective

Build an Edge Score metric combining: Directional probability (LONG/SHORT) Historical spread magnitude (Fixing2 - Fixing1) The objective is to identify hours where historical behaviour suggests a statistically and economically meaningful edge.

Recommended formula

$$\text{Edge} = (2 \times \text{Probability}_{\text{LONG}} - 1) \times \text{Median}(|\text{Spread}|)$$

where: Probability_{LONG} = percentage of days in the selected window where LONG was profitable
Median(|Spread|) = median absolute spread over the same period Median is preferred over average because it is more robust against extreme price spikes.

Preliminary Edge Interpretation (before calibration)

|Edge| < 2 No edge / No trade 2 - 5 Weak signal 5 - 10 Moderate signal 10 - 15 Strong signal 15 - 20 Very strong signal > 20 Exceptional signal

Important

The thresholds above are only initial assumptions. To create a statistically valid legend, historical data is required: hourly Fixing1 prices hourly Fixing2 prices at least 3-12 months of history The best approach is to calculate Edge historically for every hour and then determine thresholds from percentiles: P50 = neutral P75 = strong P90 = very strong P95 = exceptional This will produce market-specific thresholds instead of arbitrary values.

Suggested dashboard outputs Current heatmap (LONG/SHORT) Probability chart Edge Score chart Ranking of strongest hours for the next session